

The Island of Change

II PUC · Ch 6 · Evolution

Starring Tara 👧 & the island finches 🐦

A BAIO storybook by Narendra Sir · Biology Lecturer



Blown Away by a Storm

A wild storm blew a small flock of seed-eating birds across the sea to a distant island. Tara, a young naturalist, watched them land. "This reminds me of **Charles Darwin**," she wrote. "On his voyage aboard **HMS Beagle**, he saw small black birds on the **Galapagos Islands** — now called **Darwin's finches**."

💡 **REMEMBER**

Darwin's finches show adaptive radiation: from one seed-eating ancestor arose many forms with different beaks.



No Two Birds Alike

Tara measured every beak. Some were short, some long, some thick. "This is **variation**," she noted. "Some of it is inherited." Darwin said small variations add up over time. Later **Hugo de Vries**, studying evening primrose, said large sudden changes — **mutations** — can also create new forms in a single step; he called it **saltation**.

> 💡 **REMEMBER**
> **Darwin: evolution is gradual. De Vries: mutations are random and directionless, causing speciation.**

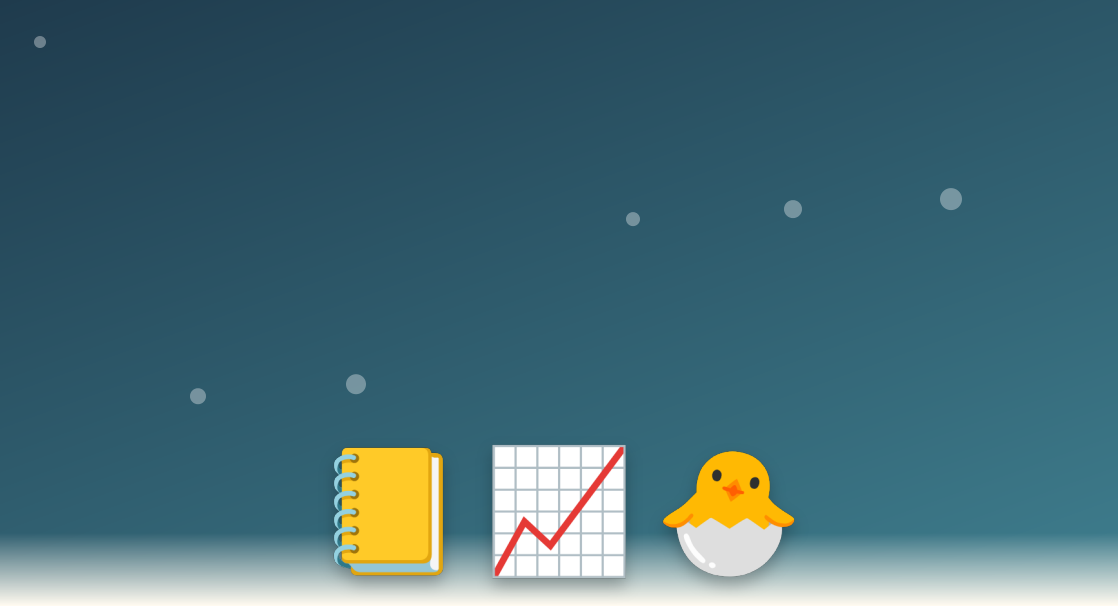


The Seeds Get Harder

Years of dry weather left only large, hard seeds. Birds with thick, strong beaks cracked them easily and raised more chicks; weak-beaked birds struggled. "Nature is choosing," Tara wrote — **natural selection**. "The winners are not the strongest fighters but those that leave more offspring. That is **fitness** — reproductive fitness." Darwin called it **survival of the fittest**.

💡 **REMEMBER**

Natural selection acts on heritable variation; traits that increase reproductive success become more common.



A Notebook of Generations

Tara recorded beak sizes for ten generations. The average beak got thicker and thicker. "A feature that helps an organism survive and reproduce in its home is an **adaptation**," she wrote. Over time, birds on different parts of the island adapted to different foods — insects, cactus, seeds — showing **adaptive radiation**, just like **Australian marsupials**.

💡 REMEMBER

Adaptive radiation: evolution of different species from one ancestor, radiating into different habitats.

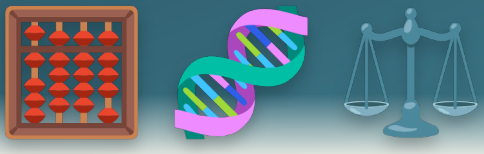


Clues in Bones and Wings

At the island museum Tara compared skeletons. The forelimbs of a whale, a bat, a cheetah and a human had the same bones but different jobs — **homologous organs**, showing **divergent evolution**. The wings of a butterfly and a bird did the same job but had different structures — **analogous organs**, showing **convergent evolution**.

💡 **REMEMBER**

Industrial melanism: in polluted England, dark moths (Biston betularia) increased because they were hidden on soot-covered trees.



Counting Alleles

"If nothing disturbs a population, allele frequencies stay constant," Tara read. That is the **Hardy-Weinberg principle**: $p^2 + 2pq + q^2 = 1$. Five factors upset this balance: **gene migration**, **genetic drift**, mutation, recombination and natural selection. "Our flock is a tiny group that started a new population — a **founder effect**!"

💡 **REMEMBER**

Genetic drift is a change in allele frequency by chance; it has a big effect in small populations.



Three Ways to Select

Tara drew three graphs of beak size. When average beaks are favoured, it is **stabilising selection**. When one extreme is favoured — like thick beaks during the drought — it is **directional selection**. When both extremes do well and the middle loses, it is **disruptive selection**. "Our island showed directional selection," she concluded.

💡 REMEMBER

Disruptive selection can split one population into two, leading to new species.



A New Island Home

Many generations later, the island birds looked different from their mainland cousins and could no longer interbreed with them. A new species had formed — **speciation**. Tara closed her notebook. "Helpful traits become common because the birds that carry them leave more young. Evolution is not a plan — it is nature's long, patient editing."

💡 **REMEMBER**

Life on Earth: first cellular forms appeared about 2000 million years ago; Homo sapiens arose in Africa about 75,000–10,000 years ago.



KEY TERMS

Words you met in this story

Tap a word to jump to its page. Can you explain each one?

Charles Darwin p.1 · HMS Beagle p.1 · Galapagos Islands p.1

Darwin's finches p.1 · variation p.2 · Hugo de Vries p.2 · mutations p.2

saltation p.2 · natural selection p.3 · fitness p.3

survival of the fittest p.3 · adaptation p.4 · adaptive radiation p.4

Australian marsupials p.4 · homologous organs p.5

divergent evolution p.5 · analogous organs p.5

convergent evolution p.5 · Hardy-Weinberg principle p.6

gene migration p.6 · genetic drift p.6 · founder effect p.6

stabilising selection p.7 · directional selection p.7

disruptive selection p.7 · speciation p.8



QUICK QUIZ

Do you remember?

1. What is natural selection?

- Humans choosing mates
- Survival and reproduction of the best-suited
- Random breeding
- Humans making genes

2. What is an adaptation?

- A disease
- A trait that aids survival
- A fossil
- A storm

3. Why do helpful traits spread over generations?

- Organisms choose them
- Their carriers leave more young
- They are learned
- Nothing changes

4. The founder effect is an example of:

- Gene migration
- Genetic drift
- Mutation
- Convergent evolution

5. Homologous organs indicate:

- Convergent evolution
- Divergent evolution
- No evolution
- Industrial melanism

Answers: 1-Survival and reproduction of the best-suited · 2-A trait that aids survival · 3-Their carriers leave more young · 4-Genetic drift · 5-Divergent evolution